

# Huanqi Wen

+8613546387712 | [wenhq7@mail2.sysu.edu.cn](mailto:wenhq7@mail2.sysu.edu.cn) | [huanqiwen.github.io](https://huanqiwen.github.io)

Guangzhou, Guangdong, 510275, China

## EDUCATION

- **Sun Yat-sen University** Guangzhou, China  
*M.S. in Mathematics* 09/2023 - Present
  - **GPA:** 89/100; 3.9/4; Top 10 %
  - **Adivisor:** Professor Lei Song
- **Sun Yat-sen University** Guangzhou, China  
*B.S. in Mathematics, Minnor in Economics* 09/2019 - 06/2023
  - **Major GPA:** 89/100; 3.9/4; Top 20 %

## RESEARCH INTERESTS

- **Pure mathematics:** toric geometry, syzygies of linear series; tropical geometry.
- **Applied mathematics:** formalization of mathematics, neuroscience and mathematics.

## RESEARCH EXPERIENCE

- **Positivity of line bundles on toric varieties** 09/2023 - present  
Study the case where the pair  $(X, L)$  of a toric variety  $X$  and a line bundle corresponding to a polytope  $P$  has the  $N_p$  properties.
  - **Related research problem:** stability of syzygy bundles.
  - **Theories/topics used:** derived category of toric varisties, resolutions of toric subvarieties, Klyachko's classification of toric vector bundles.
  - **Computer tools:** SageMath; Macauley2.
- **Minkowski sum of latitice points in lattice polytopes** 09/2022 - 09/2023  
Studied which lattice polytopes are normal, specifically whether the lattice points in an  $n + 1$  dilation of a polytope  $P$  can be written as the sum of a lattice point in  $P$  and a lattice point in an  $n$  dilation of  $P$ .
  - **Original problem:** Oda's conjecture.
  - **Studied on simplices in dimension 3 and 4:** By covering simplices with dilations, I proved that simplices with large lattice length in dimensions 3 and 4 are normal.
  - **Wrote** computer program (Python, matlab) to evaluate the normality of lattice polytopes with large lattice length.
- **Tropical geometry** 09/2023 - 01/2024  
Explored the problem "Curve Counting Problems and Tropical Geometry" and wrote a survey for a graduate-level course.

## PUBLICATIONS AND PREPRINTS

C=CONFERENCE, J=JOURNAL, P=PATENT, S=IN SUBMISSION, T=THESIS

[S.1] Lei Song, Huanqi Wen, Zhixian Zhu. (2024). **On Covering Simplices by Dilations in Dimensions 3 and 4**, arXiv preprint, manuscript submitted for publication in *Discrete & Computational Geometry*, [arXiv:2404.02495](https://arxiv.org/abs/2404.02495)

## READING SEMINAR

- **Speaker - Toric Geometry Seminar** 2024 Spring - 2024 Fall  
*Fulton's book "Introduction to Toric Varieties" and other classical papers*
- **Speaker - Syzigies Seminar** 2025 Spring  
*Irena Peeva's book "Graded Syzygies"*
- **Algebraic geometry Seminar** 2024 Spring - 2025 Spring  
*Hartshorne's book "Algebraic Geometry"*

## CONFERENCES

---

- **2024 International Summer school of Algebraic Geometry** 07/2024  
*Fudan University* [[website](#)]
- **China Southeastern Algebraic Geometry Symposium VII** 12/2024  
*Sun Yat-sen University* [[website](#)]
- **Stable reduction of algebraic curves** 05/2025  
*Xiamen University, Speaker: Qing Liu*
- **2025 SRI in algebraic geometry** 07/2025  
*Colorado state university* [[website](#)]

## EXPERIENCE

---

- **Sun Yat-sen university** 2023 Fall  
Graduate teaching assistant of linear algebra.
- **Guangdong Digital Talent Training Program** 07/2022 - 08/2022  
Conducted practical training and internships at a government digitalization department and China Telecom.
- **International Scientific Research and Development Training** 2025 Spring  
Completed the course on "Constructive Mathematics and Theoretical Computer Science," taught by Professor Vladimir Chernov.

## AWARDS AND SCHOLARSHIPS

---

- **Graduate Scholarship (Top 20 %)** 2023-2024, 2024-2025, 2025-2026  
*Sun Yat-sen University*
- **Honorable Mention, Mathematical Contest in Modeling (MCM)** 2022  
*Organized by the Consortium for Mathematics and Its Applications (COMAP)*
  - Awarded for outstanding performance in solving a real-world mathematical modeling problem as part of a three-person team in an international modeling competition.
- **Postgraduate Scholarship (Top 30 %)** 2020-2023, three times  
*Sun Yat-sen University*

## SKILLS

---

- Languages: English (IELTS: 7.0)
- Programming Languages: C++, Python
- Mathematical tools: MATLAB, SageMath, Macaulay2, LaTeX